VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

# MDI 3003 – Advanced Predictive Analysis

Sarah Blessy N
23MID 0296

**LAB Assessment 2**

Medical Diagnosis Support: Disease Classification Using Decision Trees

Industry Style Report



GitHub Repo → https://github.com/blessy228/Course---Predictive-Analysis

## 1. Executive Summary

The objective of this study was to compare different classification models and identify the model that provides the best prediction performance. Four models were evaluated: Dummy Baseline, Basic CART, Tuned & Pruned CART, and Random Forest. Model performance was assessed using ROC-AUC, PR-AUC, Sensitivity, and Balanced Accuracy obtained through cross-validation.

Among all the models, Random Forest was selected as the final model because it achieved the best overall performance. It obtained the highest ROC-AUC (0.989 ± 0.007), PR-AUC (0.988 ± 0.006), and Balanced Accuracy (0.955 ± 0.011), indicating strong classification ability and stable performance across validation folds. Although the Tuned & Pruned CART model achieved a slightly higher Sensitivity (0.947 ± 0.039) than Random Forest (0.935 ± 0.022), its overall predictive performance was lower.

The comparison also shows a clear improvement over the Basic CART model, which achieved a ROC-AUC of 0.917 ± 0.038, PR-AUC of 0.845 ± 0.061, Sensitivity of 0.894 ± 0.061, and Balanced Accuracy of 0.917 ± 0.038. The Dummy Baseline produced the weakest results, confirming that the machine learning models learned meaningful patterns from the data.

The main trade-off observed is between Sensitivity and overall classification performance. While the Tuned & Pruned CART model detected slightly more positive cases, the Random Forest model provided better discrimination, precision-recall performance, and balanced accuracy. Therefore, the small reduction in sensitivity is compensated by a significant improvement in the overall prediction quality.

Based on these results, Random Forest is recommended as the final model because it offers the most reliable and consistent performance across all evaluation metrics while maintaining high sensitivity. This makes it the most suitable model for accurate and robust classification in this study.

## 2. Intended Use and Scope

This project develops and evaluates multiple machine learning classification models, including Dummy Baseline, Basic CART, Tuned & Pruned CART, and Random Forest, to predict the target class in the given dataset. The primary purpose is to compare the performance of these models and identify the most suitable classifier based on evaluation metrics.

The intended users are students, instructors and researchers who want to study and compare classification algorithms. Predictions are generated after the trained model receives the required input features, and the output is used to classify each observation into the positive or negative class defined in the dataset.

This work is intended only for educational and experimental purposes. It demonstrates the process of training, tuning, validating, and comparing machine learning models. The models are not intended for clinical, medical, legal, financial, or other high-risk decision-making applications, and the results should not be used to make real-world decisions without further validation and domain-specific evaluation.

## 3. Dataset and Provenance

The dataset used in this study was loaded into the notebook for training and evaluating multiple classification models. The uploaded files contain the processed data used for model development and performance evaluation.

The dataset consists of a binary classification problem with one target (outcome) variable and multiple predictor variables used as input features for the machine learning models. The notebook compares the performance of Dummy Baseline, Basic CART, Tuned & Pruned CART, and Random Forest using this dataset.

## 4. Data Quality

- **Missing values:** The dataset contains **0 missing values**, indicating that no missing data handling was required.
- **Duplicate records:** No duplicate rows were found in the dataset (**0 duplicate rows**).
- **Constant columns:** No constant columns were detected, meaning all features contain useful variation for model training.
- **Data types:** The dataset consists of 30 floating-point (float64) features and 1 integer (int64) column.
- **Data plausibility:** No separate plausibility or validity checks are documented in the notebook.
- **Repeated patients/observations:** The notebook does not report whether repeated patients or repeated observations are present in the dataset.
- **Cleaning decisions:** Since there were no missing values, duplicate rows, or constant columns, no data cleaning was required before model training.
- **Audit summary:** The data audit confirms that the dataset is complete, free of duplicates, contains no constant features, and has consistent numeric data types suitable for model development

## 5. Methods

- The dataset was divided into **training** and **testing** sets for model development and final evaluation.

- The notebook performed a basic **data audit** before training. It confirmed:
  - No missing values.
  - No duplicate rows.
  - No constant columns.

- Four classification models were trained and compared:
    - Dummy Baseline
    - Basic CART (Decision Tree)
    - Tuned & Pruned CART
    - Random Forest

- The CART model was **tuned and pruned** to improve its performance and reduce overfitting.

- **Cross-validation (CV)** was used to evaluate the models and measure how well they perform on different data splits.

- The models were compared using the following evaluation metrics:
    - ROC-AUC
    - PR-AUC
    - Sensitivity
    - Balanced Accuracy

- The **Random Forest model** was selected as the final model because it achieved the best overall performance across the evaluation metrics.

- The analysis was carried out using Python in a Jupyter Notebook with machine learning libraries used for model training, tuning, and evaluation.

- The notebook follows a fixed workflow of data audit → model training → cross-validation → model comparison → final model selection before evaluating the final results.



## 6. Model Development

- **Dummy Baseline Model**
    - A Dummy Baseline model was created as the starting point for comparison.
    - This model provides a reference performance without learning meaningful patterns from the dataset.
    - It produced the lowest performance among all the models.
    - The baseline helped verify that the machine learning models performed much better than a simple prediction strategy.

- **Basic CART Model**
    - A basic Classification and Regression Tree (CART) model was trained using the training data.
    - This model learned decision rules from the input features to classify the target variable.

- The model showed good performance and performed much better than the Dummy Baseline.
- However, its overall performance was lower than the tuned CART and Random Forest models.

- **Tuned and Pruned CART Model**
  - The CART model was improved by tuning its parameters and pruning the tree.
  - Pruning reduced the size of the decision tree and removed unnecessary branches.
  - This helped reduce overfitting and improved the model's ability to perform on unseen data.
  - The tuned CART model achieved better evaluation scores than the basic CART model.
  - It also produced the highest Sensitivity, showing that it identified positive cases slightly better than the other models.

- **Random Forest Model**
  - A Random Forest classifier was trained using multiple decision trees.
  - The final prediction was obtained by combining the predictions from all the trees.
  - This model achieved the best overall performance among all the models.
  - It recorded the highest ROC-AUC, PR-AUC, and Balanced Accuracy.
  - Based on these results, Random Forest was selected as the final model.

- **Model Comparison**
  - All four models were evaluated using the same cross-validation procedure.
  - The comparison was based on four evaluation metrics:
    - ROC-AUC
    - PR-AUC
    - Sensitivity
    - Balanced Accuracy

  - The results showed a gradual improvement in performance from Dummy Baseline to Basic CART, then to Tuned & Pruned CART, and finally to Random Forest.

**Cross-Validation Results**

| Model | ROC-AUC | PR-AUC | Sensitivity | Balanced Accuracy |
|---|---|---|---|---|
| Dummy Baseline | Lowest | Lowest | Lowest | Lowest |
| Basic CART | 0.917 ± 0.038 | 0.845 ± 0.061 | 0.894 ± 0.061 | 0.917 ± 0.038 |
| Tuned & Pruned CART | Improved over Basic CART | Improved over Basic CART | 0.947 ± 0.039 | Improved over Basic CART |

| | | | | |
|---|---|---|---|---|
| Random Forest | 0.989 ± 0.007 | 0.988 ± 0.006 | 0.935 ± 0.022 | 0.955 ± 0.011 |

**Variability and Model Complexity**

- Cross-validation was used to measure the performance of each model over multiple data splits.
- The results are reported as mean ± standard deviation, showing both average performance and variability.
- Smaller standard deviation values indicate more consistent model performance.
- The Basic CART model is the simplest tree-based model.
- The Tuned & Pruned CART model is more controlled because unnecessary branches were removed.
- The Random Forest model is the most complex model because it combines many decision trees, resulting in higher accuracy and more stable predictions.
- Based on the cross-validation results, **Random Forest provided the best balance between prediction accuracy and consistency**, making it the recommended model for this study.

## 7. Final Evaluation

- The Random Forest model was selected as the final model because it achieved the best overall performance during evaluation.
- The final model achieved the following cross-validation results:
    - ROC-AUC: 0.989 ± 0.007
    - PR-AUC: 0.988 ± 0.006
    - Sensitivity: 0.935 ± 0.022
    - Balanced Accuracy: 0.955 ± 0.011
- These results indicate that the Random Forest model provides accurate and consistent predictions across different validation folds.
- The notebook compares the final model with the Dummy Baseline, Basic CART, and Tuned & Pruned CART models, and Random Forest achieved the best overall performance.

## 8. Interpretation

- The **Random Forest** model was selected as the final model because it achieved the highest overall performance among all the models.
- The **Basic CART** and **Tuned & Pruned CART** models use decision tree structures to classify the data.
- The **Tuned & Pruned CART** model is a simplified version of the basic decision tree because unnecessary branches were removed during pruning.
- The comparison results show that the **Random Forest** model performed better than the Dummy Baseline, Basic CART, and Tuned & Pruned CART models based on ROC-AUC, PR-AUC, and Balanced Accuracy.

- The notebook reports cross-validation results, showing that the Random Forest model had **low variability** (small standard deviation) across validation folds, indicating consistent performance.
- The reported results show that the selected model performs well on the available dataset. However, these results indicate predictive performance only.

## 9. Subgroup and Robustness

- The notebook evaluates all models using cross-validation to measure their performance on different data splits.
- Cross-validation results are reported as mean ± standard deviation, showing the average performance and variation across the validation folds.
- The Random Forest model showed the best and most consistent performance, with low standard deviation values for the reported evaluation metrics
- The robustness assessment in the notebook is limited to the cross-validation results used for model comparison.
- No additional robustness tests or external validation settings are reported in the uploaded files.

## 10. Limitations and Risks

- The notebook compares model performance using cross-validation only. No external validation dataset is used.
- The notebook does not report any analysis of dataset bias.
- The notebook does not discuss data distribution shift or how the model would perform on new or different datasets.
- The notebook does not describe the quality of the class labels or how the labels were created.
- The notebook does not report the dataset size, so the effect of sample size on model performance cannot be assessed.
- The notebook does not discuss privacy or how sensitive data, if any, was handled.
- The notebook does not include a discussion of possible misuse risks or inappropriate use of the trained model.
- The reported performance is based only on the available dataset and validation procedure. The model's performance on unseen real-world data is not evaluated in the uploaded files.
- Since these aspects are not documented in the notebook, they should be considered limitations of the current study and should be addressed in future work.

## 11. Deployment Considerations

- **Human oversight**
  - Predictions should be reviewed by a trained person before any important decision is made.
  - The model should support decision-making, not replace human judgment.
- **Data validation**
  - Check that all required input values are available before making a prediction.

- Verify that the data types and formats are correct.
  - Reject incomplete or invalid records.
- **Model monitoring**
  - Monitor the model's accuracy after deployment.
  - Compare new predictions with actual outcomes to detect any drop in performance.
  - Track important evaluation metrics regularly.
- **Recalibration and retraining**
  - If the model's performance decreases over time, retrain it using newer data.
  - Update the model whenever the data distribution changes significantly.
- **Rollback plan**
  - Keep a copy of the previously deployed model.
  - If the updated model performs poorly or produces unexpected results, switch back to the earlier stable version.
- **Performance testing**
  - Test the model on an independent dataset before deploying it.
  - Ensure that it performs consistently on unseen data.
- **Documentation**
  - Document the training data, model parameters, evaluation metrics, and software versions.
  - This helps reproduce the results and maintain the model.

## 12. Conclusion

- Four classification models were developed and compared:
  - Dummy Baseline
  - Basic CART
  - Tuned & Pruned CART
  - Random Forest
- The comparison showed that Random Forest achieved the best overall performance among all the evaluated models.
- The Random Forest model obtained the highest evaluation scores:
  - ROC-AUC: $0.989 \pm 0.007$
  - PR-AUC: $0.988 \pm 0.006$
  - Balanced Accuracy: $0.955 \pm 0.011$
- The Tuned & Pruned CART model achieved the highest Sensitivity ($0.947 \pm 0.039$), but its overall performance was lower than the Random Forest model.
- Based on the evaluation results, Random Forest is recommended as the final model because it provides the best balance of prediction accuracy and consistent performance.